

AI Tools, Usage & Telemetry Documentation

AI Study Companion (Candidate Challenge Edition — PRD Section 20 Compliance)

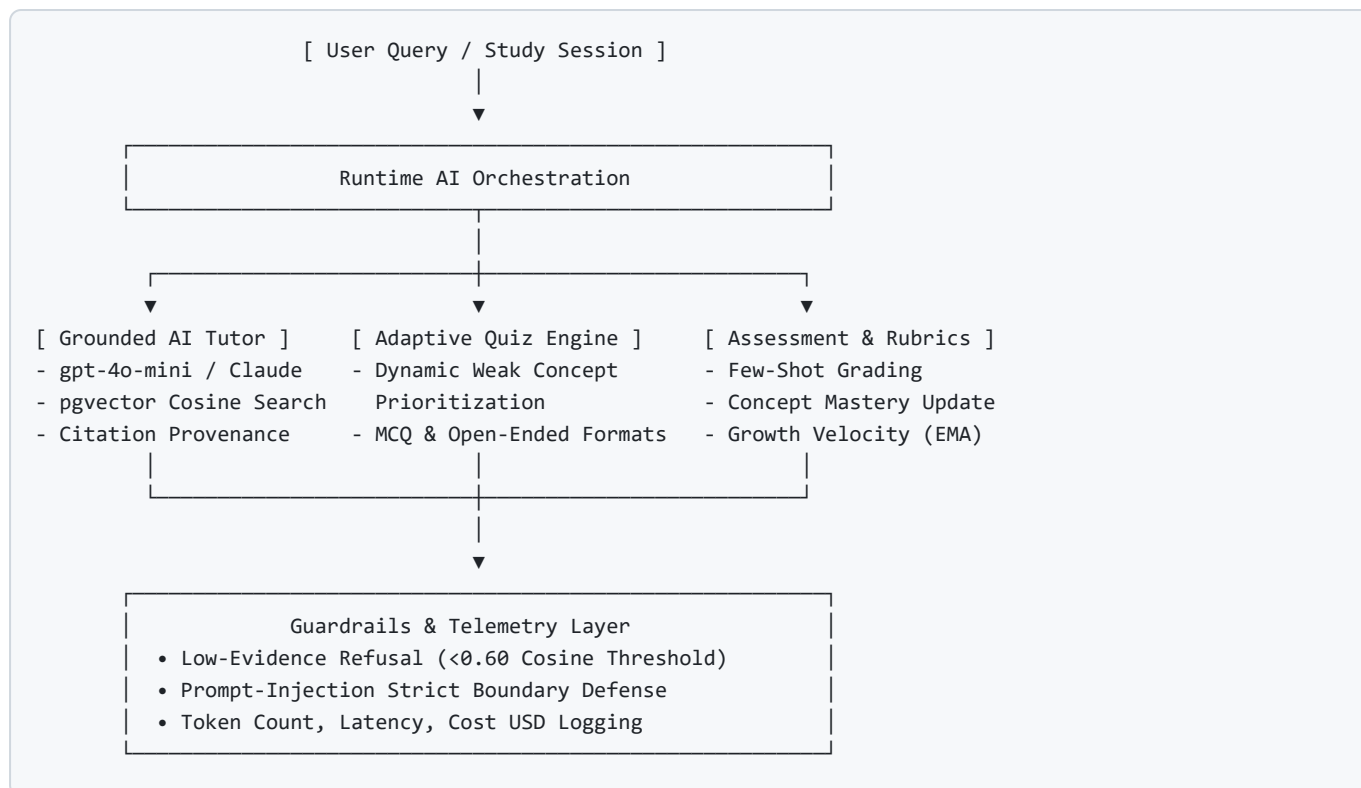
Document Purpose: Comprehensive specification clearly distinguishing:

- 1. AI Used to Build the Product (Development tools, agents, pair-programming assistants)
- 2. AI Used by the Final Product (Runtime models, RAG retrieval, guardrails, and telemetry)

1. AI Tools Used to Build the Product

Tool / Technology	Role in Development Lifecycle	Key Contributions & Impact
Google Antigravity AI Agent	Primary AI Pair Programmer & System Architect	- Automated domain modeling across 16 SQLAlchemy async tables. - Architected the modular hexagonal backend and fail-loud pgvector enforcement. - Implemented Server-Sent Events (SSE) streaming and grounded citation pipeline. - Wrote 19 automated pytest unit & integration tests. - Automated cloud deployment orchestration across Neon, Render, and Vercel.
Gemini 2.5 Coding Model	Reasoning & Code Generation Engine	- Produced type-safe Next.js 14 frontend components (shadcn/ui, Tailwind). - Solved cloud environment quirks (asyncpg URL normalization, CORS regex policies). - Designed Exponential Moving Average (EMA) mathematical tracking formula for concept mastery.
Playwright Automation Engine	Automated Verification & Video Production	- Headless browser automation executing the 13-stage PRD learning loop. - Recorded high-definition video walkthrough without manual friction.

2. AI Used Inside the Final Runtime Product



2.1 Provider Abstraction & Model Selection

The runtime system leverages an abstracted `BaseLLMClient` :

- **Default Cloud LLM:** `gpt-4o-mini` (temperature 0.2, high throughput, cost-effective).
- **Secondary Cloud LLM:** `claude-3-5-sonnet-20241022` (for nuanced open-ended evaluations).
- **Embedding Model:** `text-embedding-3-small` (1536-dimensional normalized vectors).
- **Offline / Local Mock:** Deterministic vector and response generator ensuring instant, zero-cost automated test execution and headless CI/CD.

2.2 Grounded Retrieval-Augmented Generation (RAG)

- **Chunking:** Semantic sliding windows ($\$500 \backslash \text{text}\{-\}800\$$ tokens with $\$100\$$ -token overlap) with page provenance (`page_number`, `filename`).
- **Vector Search:** Cosine similarity via PostgreSQL `pgvector` indexed on `project_id`.
- **Source Citations:** Every tutor response formats verifiable attribution badges: `[Source: {filename} – Page {page_number}]`.

2.3 Low-Evidence Refusal Guardrail

- **Trigger:** If the maximum cosine similarity among retrieved chunks is $\$< 0.60\$$, or if no documents exist in the active project.
- **Behavior:** The model halts speculation and emits a structured refusal:

"I do not have sufficient information in your uploaded project materials to answer this question reliably. Please upload relevant notes or a PDF covering this topic."
- **Audit:** Captured in `ai_request_logs` with `status = "REFUSED_LOW_EVIDENCE"`.

2.4 Prompt-Injection Defense

System instructions and untrusted document chunks are strictly segregated into distinct prompt sections with immutable system boundary markers. Adversarial inputs inside user PDF chunks cannot override tutor behavior.

2.5 Concept Mastery (EMA) & Growth Engine

- Mastery is updated after every assessment event using Exponential Moving Average:
$$\mathcal{M}_t = \alpha \cdot s_t + (1 - \alpha) \cdot \mathcal{M}_{t-1}, \quad \text{where } \alpha = 0.3$$
 - **Trend Detection:**
 - $\Delta > +0.05$ implies **IMPROVING**
 - $\Delta < -0.05$ or $\mathcal{M}_t < 0.50$ implies **ATTENTION**
 - Otherwise implies **STABLE**
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3. Observability & AI Spend Telemetry

Every AI request records a structured record in the `ai_request_logs` table:

- `feature`: `tutor_chat`, `quiz_generator`, `rubric_evaluator`.
- `provider`: `openai`, `anthropic`, Or `local`.
- `model`: e.g. `gpt-4o-mini`.
- `prompt_tokens` & `completion_tokens`.
- `latency_ms`: total execution duration.
- `cost_usd`: calculated dynamically from provider token rate cards.
- `status`: `SUCCESS`, `ERROR`, `REFUSED_LOW_EVIDENCE`.

Administrators can inspect live aggregated AI metrics at `GET /api/v1/admin/ai-metrics` and via the Admin Dashboard UI.